

Keywords: vision-language models • assistive technology • salience • low vision

Focus Where It Counts: A Salience-Driven Vision-Language Model for Low Vision Assistance

Captioning That Prioritizes Scene Elements by Human-Centered Importance, Not Photographic Coverage

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arXiv:2608.28218 • ECCV 2026

Vision-language models offer powerful scene captioning but are designed for comprehensive coverage, not for the ordered priorities of users with blindness or low vision. This paper introduces Salience-LLaVA, a salience-driven VLM trained to mention scene elements in order of their functional importance for human-centered navigation and interaction. Supported by three newly curated salience-annotated datasets, the model places the highest-priority element first in 84.3% of test images and is rated 4.2/5 by blind and low-vision users—versus 3.1/5 for standard LLaVA.

AT A GLANCE

Problem: General-purpose VLMs describe images comprehensively but do not order information by functional relevance for users with blindness or low vision.

Why it matters: BLV users often hear only the first few seconds of a caption; placing important information last renders AI assistance unreliable for navigation and safety.

Method: Curate salience-annotated datasets; fine-tune LLaVA with a combined captioning and ranking loss to produce salienceordered descriptions; prepend salience scores as soft prompts.

Result: 84.3% top-1 salience ordering accuracy (vs. 51.2% for LLaVA); 4.2/5 user helpfulness rating vs. 3.1/5.

Evidence: Three datasets, user study with BLV participants, transfer experiments on VizWiz.

The Big Picture

Assistive technology for persons with blindness or low vision (BLV) is undergoing a rapid transformation as vision-language models replace earlier rule-based systems. VLMs can describe complex scenes in natural language, answer visual questions, and identify objects—capabilities that are transformative for BLV users navigating the

world.

But VLMs are trained on captioning benchmarks that reward comprehensive coverage: a high CIDEr score requires mentioning many objects accurately. For a BLV user in a grocery store, a caption that begins with the color and pattern of the floor tiles before mentioning that the aisle is blocked by a forklift is not just unhelpful—it is dangerous.

This paper identifies the ordering problem as a first-class challenge in assistive VLM design and proposes a framework that addresses it through salience-aware training.

Why Existing Approaches Fall Short

General-purpose captioning: Models trained on MSCOCO and similar benchmarks describe images from top-left to bottom-right, or by object size/saliency in the photographic sense. Neither corresponds to functional importance for BLV navigation or interaction.

VQA-based assistance: Answering specific questions (“Is the crosswalk light green?”) requires the user to know what to ask. Open-ended scene description is essential for unknown environments.

Prior assistive captioning: Work on VizWiz-focused models improves answer accuracy for BLV-generated images but does not address the ordering of information in free-form descriptions.

The key missing component is a notion of *salience* grounded in human-centered priorities rather than photographic composition.

Core Mechanism

The Salience-LLaVA framework has two components:

Salience scoring module: A lightweight MLP head added to the VLM’s visual encoder that predicts a scalar salience score $s_i \in [0, 1]$ for each detected bounding box o_i . The module is trained with a point-wise ranking loss against human-annotated salience ranks.

Salience-conditioned caption generation: The caption generator receives the ranked object list $[o_{\pi(1)}, o_{\pi(2)}, \dots]$ (sorted by predicted salience) as soft prompt tokens prepended to the image features. The LM is then trained to generate captions that mention objects in the order they appear in the ranked list.

At inference, salience scoring and caption generation run in sequence: score all detected objects, sort by salience, generate a caption conditioned on the sorted list.

Technical Formulation

Let $O = \{o_1, \dots, o_n\}$ be the detected objects with ground-truth salience scores s_i^* . The salience head predicts $\hat{s}_i$ and

the ranking loss penalizes misordered pairs:

$$\mathcal{L}_{\text{rank}} = \sum_{i,j: s_i^* > s_j^*} \max(0, \hat{s}_j - \hat{s}_i + \delta)$$

with margin $\delta = 0.1$.

The total training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(y, \hat{y}) + \lambda \mathcal{L}_{\text{rank}}$$

where $\mathcal{L}_{\text{CE}}$ is cross-entropy on the caption sequence y and $\lambda = 0.3$.

The implicit ordering $\hat{\pi}$ extracted from the generated caption is defined by the first mention position of each object:

$$\hat{\pi}(i) = \min\{t : o_i \text{ is mentioned at token position } t\}$$

The ranking loss targets the ordering induced by $\hat{s}$ matching s^* .

Learning or Inference Procedure

Dataset construction:

1. **Salience COCO:** 30k MSCOCO images; annotators rank all detected objects by importance for a BLV person navigating the scene.
2. **Salience Flickr:** 10k Flickr images with the same protocol.
3. **Salience VizWiz:** 15k images from BLV users' own photographs; salience ranks derived from crowd-sourced question answers.

Training: Fine-tune LLaVA-1.6 on Salience COCO and Salience Flickr for 3 epochs with $\lambda = 0.3$; evaluate zero-shot transfer to Salience VizWiz.

Inference: Run object detection (DINO-v2), score salience, sort, generate caption with sorted soft prompt. Total overhead: +38 ms per image on an A100.

What the Guarantee Says

The framework guarantees salience ordering in expectation over the training distribution: at convergence, the expected rank of the highest-salience object's first mention position approaches 1. The guarantee is probabilistic—individual captions may reorder objects due to LM fluency constraints—but the aggregate improvement is statistically significant ($p < 0.001$).

Experimental Findings

- **Top-1 salience ordering:** Salience-LLaVA 84.3% vs. LLaVA-1.6 51.2%, GPT-4V-captioning 58.7%.
- **VizWiz transfer:** Salience ordering F1 +12.3 pp over LLaVA-1.6 without VizWiz fine-tuning.

- **Caption quality:** SPICE +2.1, CIDEr +5.3 over LLaVA-1.6, confirming salience ordering does not hurt coverage.
- **User study (n=24 BLV participants):** Helpfulness 4.2/5 vs. 3.1/5 for LLaVA; 20/24 preferred Salience-LLaVA for navigation tasks; most cited "important info came first."

Ablations and Interpretation

Soft prompt vs. hard ordering: Providing the ranked object list as a hard prefix (forcing the caption to start with each object in order) achieves 100% top-1 ordering but reduces fluency (SPICE −8.4). Soft prompting achieves 84.3% ordering with only +2.1 SPICE gain.

Salience source: Using ground-truth salience ranks at inference yields 92.1% top-1 ordering, establishing an upper bound; the predicted salience scores are 93% rank-correlated with human rankings.

λ ablation: $\lambda = 0$ (no ranking loss) reduces top-1 ordering to 54.1%; $\lambda = 1.0$ improves ordering to 87.2% but reduces CIDEr by 11 points; $\lambda = 0.3$ balances both.

Domain specificity: Salience annotations vary across domains: in traffic scenes, moving objects (cars, cyclists) are highest salience; in indoor scenes, social agents (people) dominate. The model generalizes well within COCO categories but may require domain-specific annotation for specialized environments.

Reference

Jiazhaoh Liang, Hao Huang, Shuaihang Yuan, et al. **Focus Where It Counts: A Salience-Driven Vision-Language Model for Low Vision Assistance.** arXiv:2608.28218, ECCV 2026. <https://arxiv.org/abs/2608.28218>